Thomas Janssoone

Research Scientist / ML Engineer – Speech, Multimodal AI & LLM Evaluation
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scholar.google.fr/citations?user=qkOKEfMAAAAJ

Summary

Research scientist and ML engineer with 15+ years in speech, multimodal and clinical AI, and a PhD in affective computing (Sorbonne Université). I build real-time voice pipelines (STT, LLM, TTS) for social robots and emergency-call triage, evaluate and fine-tune LLMs, and have pre-trained long-context genomic foundation models (128 kbp). Measurement-first: pre-registered ablations, bootstrap confidence intervals, interpretable models. 15 peer-reviewed publications (NeurIPS and MICCAI workshops, ICMI, IVA, HHAI).

Skills

- **Speech & audio:** ASR (Whisper, Parakeet, Qwen3-ASR, Voxtral, Wav2Vec), speaker diarization (Sortformer, pyannote, DiariZen), TTS (Parler-TTS, Coqui, Matcha-TTS), VAD, turn-taking, full-duplex models (Moshi, PersonaPlex), real-time streaming (WebSockets, WebRTC)
- **LLMs & agents:** LLM evaluation (LLM-as-a-judge, function-calling accuracy), fine-tuning, RAG, tool calling, activation steering, multi-agent systems, local/on-premises deployment (Gemma)
- **Machine learning:** deep learning, multimodal fusion, foundation-model pre-training (long context, Mamba/SSM), sequence and time-series modelling, survival analysis, interpretability (SHAP, LIME), statistical evaluation
- **Engineering:** Python, PyTorch, Hugging Face, NVIDIA NeMo, scikit-learn, pandas, Polars, NumPy, SQL, C++, ONNX, Docker, GitLab CI, SLURM, MLflow, Weights & Biases, DVC, Linux
- **Languages:** French (native), English (fluent, C1/C2)

Experience

AI Researcher, Plant Genomic Foundation Models, Living Models, Paris — Apr 2026 – Sep 2026

- Co-developed BOTANIC-1, plant-specialist genomic language models pre-trained to a 128 kbp context (twice the previous plant-model record) on a 320-genome corpus; co-author of the bioRxiv preprint.
- Ran pre-training on a 10-node, 120-GPU SLURM cluster (University of Florida), orchestrating training runs with LLM agents.
- Built the agentic use case: a local Gemma agent that keeps proprietary breeding genotypes on-premises and calls BOTANIC-1 as a zero-shot variant-effect scorer.
- Led the NeurIPS 2026 Competition Track submission for PlantGenomeBench, a four-track evaluation framework for plant genomic models, with the University of Florida.

AI Researcher, Medical Speech Analysis, e-sensia, Paris — Sep 2025 – Feb 2026

- Architected low-latency, real-time audio streaming pipelines (WebSockets, WebRTC) for emergency medical triage on live calls.
- Built real-time inference loops with Whisper, Wav2Vec and EnCodec to extract vocal biomarkers and social signals within latency budgets; evaluated MedGemma for medical use cases.
- Wrote the ESYNAPSE grant, selected as a France 2030 "Pionniers de l'IA" laureate (up to €10M); prepared Research Tax Credit (CIR) applications.

Multimodal Machine Learning Expert, Social Robotics, Enchanted Tools, Paris — Apr 2022 – Aug 2025

- Engineered the real-time voice interaction stack of the Mirokai robot (STT, LLM/VLM dialogue, TTS) with sub-second loops, voice activity detection, turn detection and barge-in.
- Deployed WebSockets/WebRTC transport for bidirectional audio and video between embedded hardware (NVIDIA Jetson) and GPU backends.
- Benchmarked and fine-tuned LLMs and SLMs (prompting, fine-tuning, RAG, tool calling for robot actions) on real and synthetic data, with daily metrics (BLEU, function-calling accuracy, LLM-as-a-judge).
- Fine-tuned, voice-cloned and benchmarked TTS models (Parler-TTS, Coqui, Matcha-TTS) in English, French and Japanese on Jetson Orin NX/AGX, trading off speed, expressivity and tone fidelity.
- Co-supervised a PhD; co-organised the ACE workshops at ICMI 2023 and IVA 2024; program committee of IVA 2023, IVA 2024 and ECAI 2024.

Postdoctoral Researcher, Interpretable Social AI, Inria (COML team), Paris — Apr 2021 – Dec 2021

- Drove the research direction of an interpretable, dyadic multimodal model estimating conversational rapport with SHAP (HHAI 2022).
- Built a Python library for multimodal social-signal extraction and an open-source AI-assisted annotation tool for

children's communication.
- Explored generative conversational strategies (RL-GAN, BlenderBot, GPT-2). Best Reviewer award, ICMI 2021.

Research Scientist, Healthcare Machine Learning, Sêmeia, Paris Apr 2018 – Jan 2021

- Led a team of 2 data scientists working on SNIIRAM, the French national health database (66 million people).
- Combined weak signals (pharmacy purchases, care context) to identify over 90% of patients at risk of early treatment discontinuation.
- Built a Python library of survival-analysis and deep-learning models, with SHAP/LIME explanations for medical staff; published at NeurIPS 2018 workshops and MICCAI 2018 (PRIME).

Founder, Smart Farming System, Paris Jun 2018 – Nov 2022

- Founded a start-up designing AI-controlled vertical farms; first product, pop-farm, grows herbs and fruits inside supermarkets.

Earlier career: R&D Engineer, Grenoble 2010 – 2014

- LIG, Human-Computer Interaction group (2013–2014): 3D gesture guidance for ambient lighting control with Kinect and augmented reality (AVI 2016).
- TIMC-IMAG, CAMI team, with Imactis and SurgiQual Institute (2010–2012): CT-guided navigation and computer-assisted surgery software.

Independent Research (2026)

- **Hormonal Agents.** LLM agents with hormone-like internal dynamics. Pre-registered ablations across the prisoner's dilemma, coding tasks and τ^2-bench showed the gains came from prompt text, not the dynamics; a per-partner state steering the model's activations is the one mechanism that beats replay controls. Paper in preparation.
- **Concilium.** A council of small language models that deliberate and vote, one model fine-tuned per member of the French Assemblée nationale to predict votes and speeches, extended to the Conseil de Paris, with tone steering and expressive speech synthesis. Paper in preparation.
- **Local ASR and diarization benchmark.** 11 ASR systems and 3 diarizers on 162 English and French datasets (synthetic noisy multi-speaker scenes), with error analysis beyond WER (cpWER, DER) and hierarchical bootstrap confidence intervals. Found that speaker attribution, not transcription, causes 38–63% of errors; fixed a streaming bug in transcribe.cpp (WER 54.9% to 31.7%).

Education

PhD in Computer Science, Affective Computing, Sorbonne Université and Télécom Paris 2014 – 2018

- At ISIR (Sorbonne) and Télécom Paris: temporal association rules for multimodal social signals (face, head, prosody, turn-taking), used to animate virtual agents expressing stances such as dominance or friendliness. Visiting researcher, USC Institute for Creative Technologies, Los Angeles (2017).

Engineering degree (MSc), Applied Mathematics and Computer Science, Ensimag, Grenoble 2006 – 2010

- Specialisation in image and virtual reality; exchange semester at Eindhoven University of Technology (expert systems, medical imaging).

Selected Publications (full list of 15 on Google Scholar)

- A. Barozet, . . . , T. Janssoone, et al. BOTANIC-1: a series of long-context plant genomic foundation models in the agentic era. bioRxiv, 2026.
- R. Vallée, . . . , T. Janssoone. Social logistics robot in healthcare facilities: presenting the robot as a subject versus an object. IEEE ICCMI, 2025.
- G. Grimberg, T. Janssoone, C. Clavel, J. Cassell. Estimating rapport in conversations: an interpretable and dyadic multi-modal approach. HHAI, 2022.
- T. Janssoone et al. Machine learning on electronic health records: models and features usages to predict medication non-adherence. ML4H Workshop, NeurIPS, 2018.
- T. Janssoone et al. Predictive patient care: survival model to prevent medication non-adherence. PRIME Workshop, MICCAI, 2018.
- T. Janssoone, C. Clavel, K. Bailly, G. Richard. Using temporal association rules for the synthesis of embodied conversational agents with a specific stance. IVA, 2016.